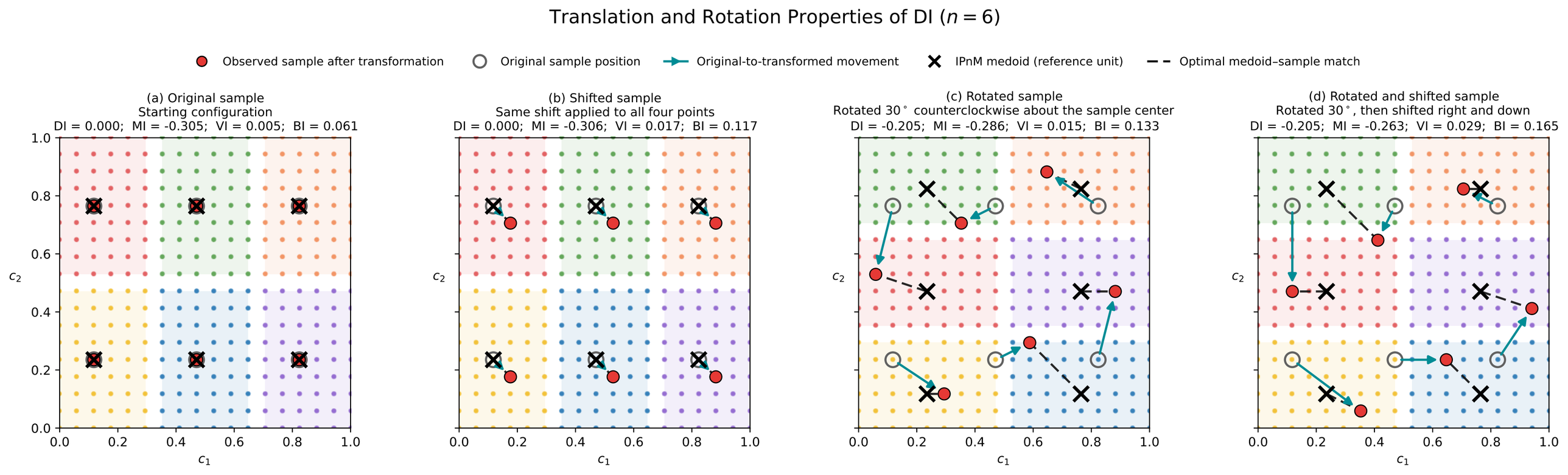

Translation and Rotation Properties of DI ($n = 6$)
Observed sample after transformation
Original sample position
Original-to-transformed movement
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Original sample
Starting configuration
DI = 0.000; MI = -0.305; VI = 0.005; BI = 0.061
(b) Shifted sample
Same shift applied to all four points
DI = 0.000; MI = -0.306; VI = 0.017; BI = 0.117
(c) Rotated sample
Rotated 30° counterclockwise about the sample center
DI = -0.205; MI = -0.286; VI = 0.015; BI = 0.133
(d) Rotated and shifted sample
Rotated 30°, then shifted right and down
DI = -0.205; MI = -0.263; VI = 0.029; BI = 0.165
$c_1$
$c_2$
0.0
0.2
0.4
0.6
0.8
1.0